Git Hub

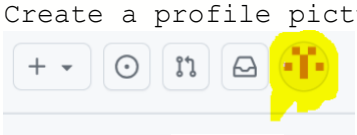
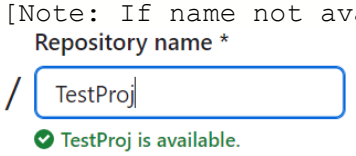
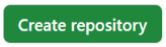
By understanding how to manage different versions of project files, team members can collaborate more efficiently and avoid conflicts.

1. Learning how to **create repositories** and **switch between file states in Git** allows for better organisation and tracking of changes throughout the project's development.

2. Understanding how to utilise **Git's branching feature** enables developers to experiment with new ideas without affecting the main project, providing a safe space for innovation and exploration.

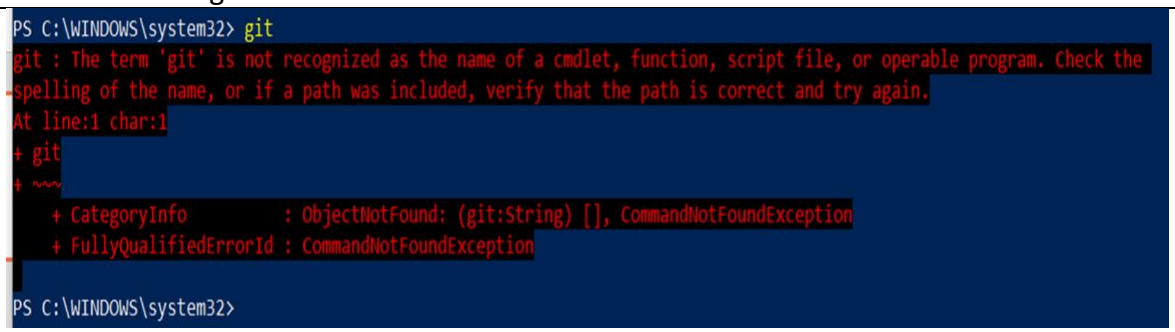
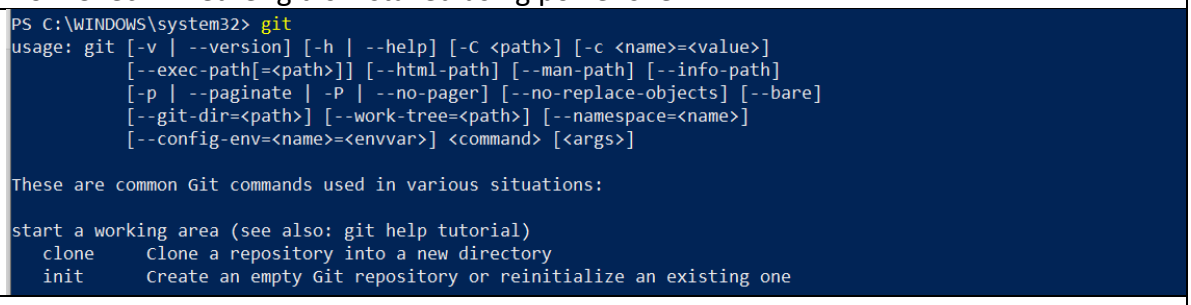
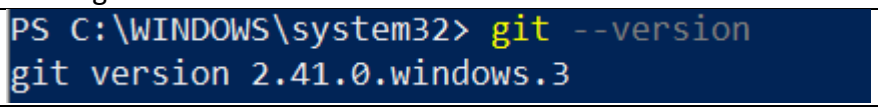
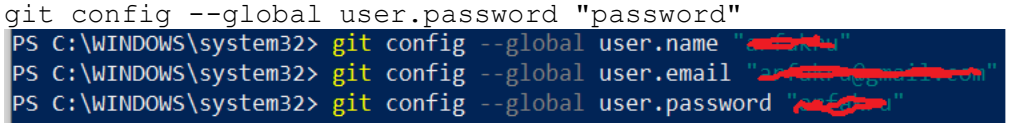
3. Git provides a version control system that allows team members to **easily revert back to previous versions** of files if needed, ensuring that no important changes are lost or overwritten.

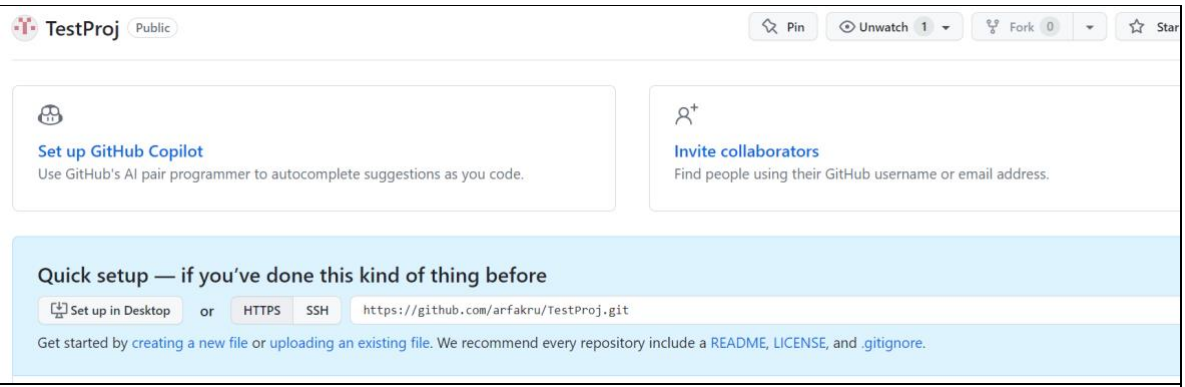
Lab Task – 1 **Creating repository in Git Hub**

a.	Create an account in Git Hub and login into Git Hub
	https://github.com/login
b.	Create a new repository for your project in Git Hub
	Create a profile picture and chose your repositories  Click new button and give repository name as "TestProj" [Note: If name not available, give other suitable name for the repo]  Then click create repository button 

Lab Task – 2 **Working with Git**

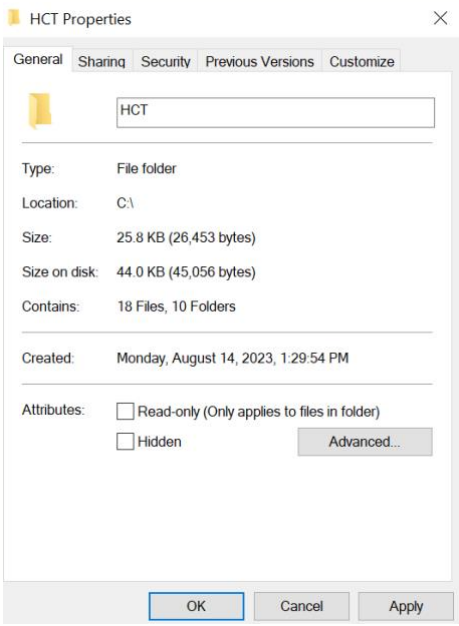
Open the windows power shell(or command prompt) and type the following commands

a.	Check whether git is installed
	 Note: if it is not installed, you will get message like in the picture
b.	Download git if it is not installed
	Download Git https://git-scm.com/downloads Install the git from the downloaded file with default options
c.	Now Check whether git is installed using power shell
	
d	Check git version
	
e.	Configure git with username and password
	git config --global user.name "username" git config --global user.email "emailid@xyz.com" git config --global user.password "password"  Note: username, pwd and email of GitHub
f.	Goto Git Hub and copy the URL of your repo



h. Clone git hub repo to Git

Create a folder called HCT using windows or power shell. In this folder, you are going to bring the repo of Git Hub
Note: the folder should have read and write permission when you clone a repo. Right click the folder—properties---general---uncheck the read-only.



Write the following command in power shel to clone.

```
PS C:\HCT> git clone https://github.com/arfakru/TestProj.git
Cloning into 'TestProj'...
warning: You appear to have cloned an empty repository.
```

i. After cloning, Check the content of repo in the Git

```
PS C:\HCT> ls

    Directory: C:\HCT


Mode                LastWriteTime         Length Name
----                -------------         ------ ----
d-----        8/14/2023   1:31 PM                TestProj
```

Note: By Default, a folder will be present in the repo name

j. Create a python file in the Testproj folder

add.ipynb

File Edit View Insert Runtime Tools Help All changes saved

+ Code + Text

```
a=1
b=2
print(a+b)
```

```
3
```

Note: Save and Download the file in Testproj folder and check it in windows

This PC > Windows (C:) > HCT > TestProj

Name	Date modified	Type
add	8/14/2023 2:47 PM	Jupyter Source File

Note: we have save the file in local repo only in the Git

k. Check the add file in the local repo

```
PS C:\HCT> cd TestProj
PS C:\HCT\TestProj> ls

    Directory: C:\HCT\TestProj


Mode                LastWriteTime         Length Name
----                -------------         ------ ----
-a----        8/14/2023   2:47 PM            768 add.ipynb
```

l. Convert the add file for tracking by using add command and check the status

```
PS C:\HCT\TestProj> git add .\add.ipynb
warning: in the working copy of 'add.ipynb', LF will be replaced by CRLF the next time Git touches it
PS C:\HCT\TestProj> git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   add.ipynb
```

Note: The file is not yet commited

m Save the file to the local repo

```
PS C:\HCT\TestProj> git commit -m "Add python file is added"
[main (root-commit) 3aefef5] Add python file is added
 1 file changed, 43 insertions(+)
 create mode 100644 add.ipynb
PS C:\HCT\TestProj>
```

Check the status of the file in the loca repo

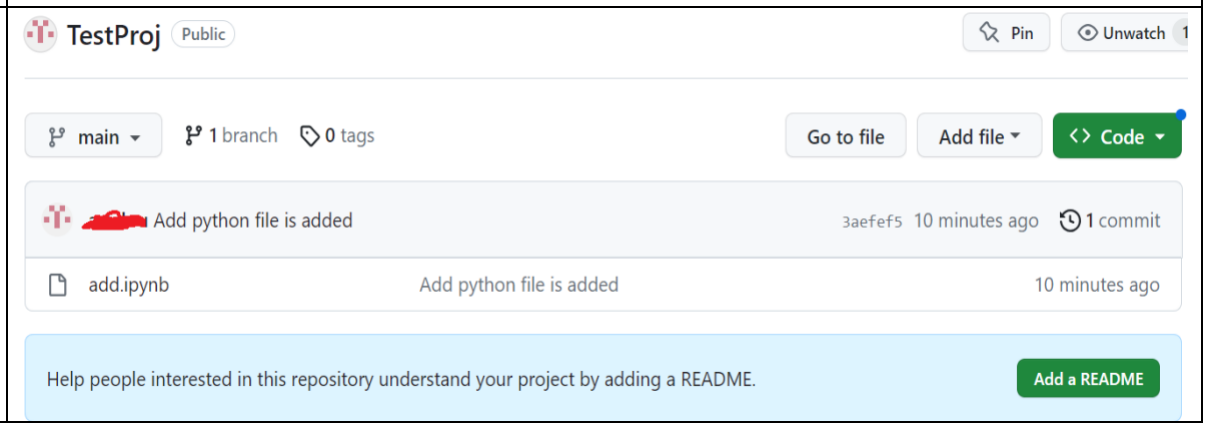
```
PS C:\HCT\TestProj> git status
On branch main
Your branch is based on 'origin/main', but the upstream is gone.
  (use "git branch --unset-upstream" to fixup)

nothing to commit, working tree clean
PS C:\HCT\TestProj>
```

Push the add file from local repo(Git) to remote repo(Git Hub)

```
PS C:\HCT\TestProj> git push
info: please complete authentication in your browser...
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 558 bytes | 558.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/[redacted]/TestProj.git
 * [new branch]      main -> main
PS C:\HCT\TestProj>
```

Now check it in Git Hub



Preview | Code | Blame 43 lines (43 loc) · 768 Bytes

In [1]:
```
a=1
b=2
print(a+b)
```

```
3
```

Consider this Testproj in Git Hub is accessed by your project team members and they edit the file as shown below in the Git Hub.

Open the add.ipynb from the colab through Git Hub repo

Examples | Recent | Google Drive | GitHub | Upload

Enter a GitHub URL or search by organisation or user Include private repos

https://github.com/[redacted]/TestProj/blob/main/add.ipynb

Repository: arfakru/TestProj Branch: main

Path

add.ipynb

Note : Choose add.ipynb to add to the colab

Modify the code as a=1 to a=10

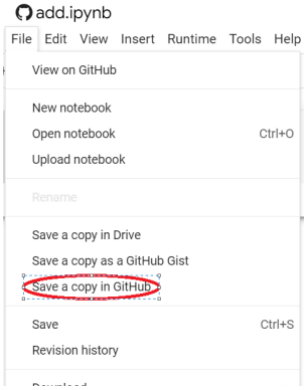
add.ipynb

File Edit View Insert Runtime Tools Help Cannot save changes

+ Code + Text Copy to Drive

```
a=10
b=2
print(a+b)
```

```
12
```

	Save the file in the GitHub
	In the colab, file—save a copy in Git Hub 
	Check the add file in the local repo

```
PS C:\HCT\TestProj> ls


    Directory: C:\HCT\TestProj


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----         8/14/2023   2:47 PM            768 add.ipynb


PS C:\HCT\TestProj> notepad.exe ./add.ipynb
```

Since it is a Jupiter file and it will come in different format. Just check the value of a in source

```
"source": [
  "a=1\n",
  "b=2\n",
  "print(a+b)"
]
```

You can see still, a=1

Pull the updated files from the remote repo(Git Hub) to local repo(Git)

```
PS C:\HCT\TestProj> git pull
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
Unpacking objects: 100% (3/3), 325 bytes | 10.00 KiB/s, done.
From https://github.com/arfakru/TestProj
   3aefef5..76a5ca7  main       -> origin/main
Updating 3aefef5..76a5ca7
Fast-forward
 add.ipynb | 6 +++---
 1 file changed, 3 insertions(+), 3 deletions(-)
PS C:\HCT\TestProj>
```

Now check the local repo whether files are updated from remote repo

```
PS C:\HCT\TestProj> notepad.exe ./add.ipynb
```

```
],
"source": [
  "a=10\n",
  "b=2\n",
  "print(a+b)"
]
```

So, you see the local repo is updated since a=10

Suppose you have change the a value to15 and some other person has change the a value to 22 in the remote repo. In that case, if you perform git pull, then merge conflict occurs.

Change the a=15 in the loca repo

1) Open the file

```
PS C:\HCT\TestProj> notepad.exe ./add.ipynb
```

2) Change the a to 15 and save it

```
"source": [
  "a=15\n",
  "b=2\n",
  "print(a+b)"
]
```

3) Commit it to the local repo

```
PS C:\HCT\TestProj> git commit -am "Add python file is added"
[main fd4926d] Add python file is added
 1 file changed, 1 insertion(+), 1 deletion(-)
```

Note: use – am options

Modify a=22 in Git Hub

1) Open add.ipynb in colab using Git Hub and modify a=22

Examples | Recent | Google Drive | GitHub | Upload

Enter a GitHub URL or search by organisation or user

☐ Include private repos

https://github.com/[redacted]/TestProj/blob/main/add.ipynb

Repository: arfakru/TestProj

Branch: main

Path

add.ipynb

2) Save it in the Git Hub

Copy to GitHub

Repository: [redacted]/TestProj

Branch: main

File path

add.ipynb

Commit message

modified the value of a to 22

☐ Include a link to Colaboratory

Cancel OK

3) Check it in the Git Hub

Preview | Code | Blame | 43 lines (43 loc) · 770 Bytes

In [1]:
```
a=22
b=2
print(a+b)
```

```
12
```

Now pull the file from remote repo to local repo

```
PS C:\HCT\TestProj> git pull
Auto-merging add.ipynb
CONFLICT (content): Merge conflict in add.ipynb
Automatic merge failed; fix conflicts and then commit the result.
```

Merge conflicts occurs because local repo contains a=15 but remote repo contains a=22

Open add file from the loca repo

```
PS C:\HCT\TestProj> notepad.exe ./add.ipynb
```

```
      "source": [
<<<<<<< HEAD
        "a=15\n",
=======
        "a=22\n",
>>>>>>> 19af11fa44b78a307a99863ff202efd853bc16e8
        "b=2\n",
        "print(a+b)"
      ]
```

Solve the merge conflict

You can keep a=15 or a=22 or assign any new value to a. So, I'm making a=20.
Remove the other tags and update as shown in the figure

```
 "source": [

   "a=20\n",
   "b=2\n",
   "print(a+b)"
 ]
```

Save the file

Save to the local repo

```
PS C:\HCT\TestProj> git commit -am "Modified a to 20"
[main 6dfbf12] Modified a to 20
```

Push the local repo to remote repo

```
PS C:\HCT\TestProj> git push
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 4 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (9/9), 875 bytes | 291.00 KiB/s, done.
Total 9 (delta 3), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (3/3), completed with 1 local objec
To https://github.com/arfakru/TestProj.git
   19af11f..71084e7  main -> main
PS C:\HCT\TestProj>
```

Confirm whether a=20 in the remote repo(Git Hub)

Preview | Code | Blame 44 lines (43 loc) · 77

In [1]:
```
a=20
b=2
print(a+b)
```

```
12
```

Perform Branching

Create a branch to modify a+b to a*b in add.ipynb

Check whether any branch exists

```
PS C:\HCT\TestProj> git branch
* main
```

Create a new branch called “NewBranch” and find the status

```
PS C:\HCT\TestProj> git checkout -b "NewBranch"
Switched to a new branch 'NewBranch'
PS C:\HCT\TestProj> git branch
* NewBranch
  main
```

- Indicates Newbranch is in active stage

Modify the content a+b to a*b

```
PS C:\HCT\TestProj> notepad.exe ./add.ipynb
```

```
],
"source": [

  "a=20\n",
  "b=2\n",
  "print(a*b)"
]
```

Save the file

Commit the file to loca repo

```
PS C:\HCT\TestProj> notepad.exe ./add.ipynb
PS C:\HCT\TestProj> git commit -am "Switch to multipl
On branch NewBranch
nothing to commit, working tree clean
PS C:\HCT\TestProj>
```